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DATA DESCRIPTOR

Multi-platform satellite-derived products during the 2025 Etna eruption

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Earth Observation data are playing an increasingly central role in volcanology, enabling high-resolution assessments of the timing, magnitude, and explosivity of eruptive events. A comprehensive suite of satellite-derived products is provided here, documenting the February 2025 eruption of Mt. Etna (Italy), the first eruption fully monitored also by the third-generation Meteosat satellite, providing unprecedented mid-infrared spatial and temporal resolution. Daily Planet's acquisitions enabled consistent monitoring of lava flow evolution, while a post-eruptive Pléiades triplet allowed for the development of an updated digital surface model and precise estimation of the deposits thickness. The dataset includes: time averaged discharge rates (TADR) from MODIS, SEVIRI, VIIRS, and FCI sensors; lava flow areal expansion from Skysat and PlanetScope imagery; deposit thickness from DSMs differencing; SO₂ mass flux from TROPOMI; a 5 m-resolution DSM from Pléiades imagery (March 6th, 2025). Combined analysis of SO₂ emissions and TADR improves understanding of magma supply and eruptive output. Furthermore, the products provided offer critical data for analyzing multi-temporal morphological changes and hazard assessment studies.

Background & Summary

In February 2025, a new eruptive phase both effusive and explosive started on Mt Etna (Italy). The explosive-type activity began on February 6th, 2025 with a modest and episodic Strombolian activity produced by a vent located in the western sector of the South-East Crater (SEC), while on February 8th, a new effusive phase began, fed by an eruptive fissure opened at the base of the Bocca Nuova (BN) crater at an elevation of ~3,050 meters above sea level (a.s.l.). The lava flow advanced rather rapidly southwestward over a snow-covered surface, reaching a quote of ~2,150 meters a.s.l. on February 11th and 1,960 meters a.s.l. on February 13th. The interaction between the lava and the snowpack caused sudden explosions projecting incandescent fragments of lava up to several tens of meters away. Weak explosive activity was also detected from a vent located between the SEC and the BN (Fig. 1). The explosive activity produced mild volcanic ash emissions, which were rapidly dispersed by the winds and deposited in the immediate proximity of the volcano summit. On February 12th, 14th and 16th, the analysis of images from the INGV-OE video surveillance system, together with field observations, showed an increase in volcanic ash emission with an eruptive column not exceeding 5,500 meters a.s.l. After a reduction in effusive activity at the vents below the BN and interruption of the SEC's explosive activity, lava emission resumed from February 22th, accompanied by the return of Strombolian activity at the SEC on February 23th. This activity continued until the morning of the 25th, when Strombolian activity at the SEC again came to a halt and the effusive rate decreased significantly. In the evening of February 27th, the SEC reactivated with Strombolian explosions, followed by the resumption of effusive activity from the vent at the base of BN.

A small lava overflow was also observed from the southern rim of SEC on February 24th and 28th. Moreover, the presence of fissures, cracks and a new pit crater on the southern flank of the BN were detected, with an intermittent emission of steam.

The effusive activity beneath the BN generated multiple lava branches, which expanded on the upper southwestern flank, thus enlarging the upper part of the lava field that had been forming since February 8th (Fig. 1). In the evening of February 28th, explosive activity at SEC gradually decreased and came to a halt during the night.

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Fig. 1 (a) Summit area of Etna volcano, showing the main craters, the eruptive vents and fissures (in red) and the lava flows (yellow area) emitted during the February 2025 eruption. (b) Location of Etna in Sicily, Italy. (c) Etna’s eruptive activity at sunset on February 16th, 2025. Photograph by Francesco Zuccarello.

	Spatial Resolution	Temporal Resolution	Product
SEVIRI	3 km (nadir)	every 15' every 5' (RSS)	TADR, Lava Volume
FCI	1 km (nadir)	every 10'	TADR, Lava Volume
VIIRS	375 m (nadir) 750 m (nadir)	twice daily	TADR, Lava volume
MODIS	1 km (nadir)	twice daily	TADR, Lava volume
Skysat	0.5 m	on demand	Lava flow fields
PlanetScope	3 m	daily	Lava flow fields
Pléiades	0.5 m	on demand	DSM, 3D map of volcanic deposits
TROPOMI	5.5 × 3.5 km ²	daily	SO ₂ mass

Table 1. Satellite imagery used in this study alongside the spatial and temporal resolution and the main derived products. TADR = Time-Averaged Discharge Rate, DSM = Digital Surface Model.

Flashes at the effusive fissure below the BN were still visible at dawn on March 1st, and a faint active lava flow was observed until March 2nd.

This eruption is a magnificent example of how a multitude of information from satellite imagery has made it possible to reconstruct and quantify in real time the evolution of the different phases of eruptive activity. Here we provide the different products we retrieved, including the expansion of the lava field, the average eruptive rates and volumes emitted, as well as the thickness of volcanic deposits (Table 1).

A brief description of the main characteristics of the different satellite imagery used (Table 1) is reported below.

The *Spinning Enhanced Visible and Infrared Imager* (SEVIRI) is the primary payload of the Meteosat Second Generation geostationary satellites. The mission, which was launched in 2002, acquires data on 11 spectral channels at 3 km spatial resolution at nadir, with an additional panchromatic high-resolution channel (1 km spatial resolution at nadir). The primary acquisition service provides a full disk image of the Earth every 15 minutes, with a sub-satellite longitude of 0°. A Rapid Scan Service (RSS) is also available, that only scans the upper third of the full disk and provides an image every 5 minutes. The RSS satellite is typically located at 9.5° longitude east; 8 of the 11 standard channels are in the infrared and one of these (Channel 4, 3.9 μm central wavelength) can be used to monitor and characterize high temperature events, such as forest fires and volcanic eruptions, especially paired with Channel 10 (10.8 μm central wavelength).

The *Flexible Combined Imager* (FCI) is the successor of the SEVIRI sensor onboard the Meteosat Third Generation geostationary satellites. The mission was launched in December 2023 and acquires data over 16 spectral channels with a spatial resolution that varies from 0.5 km to 2 km at nadir. The infrared channels at 3.8 μm and 10.5 μm are particularly important for volcanic monitoring, as they are captured in High Resolution mode with a spatial resolution of 1 km at nadir and a temporal frequency of 10 minutes.

The *Visible Infrared Imaging Radiometer Suite* (VIIRS) instrument observes and collects global satellite observations that span the visible and infrared wavelengths across land, ocean, and atmosphere. It has 22 channels ranging from 0.41 μm to 12.01 μm. Five of these channels are high-resolution image bands or I-bands (375 m spatial resolution), and sixteen serve as moderate-resolution bands or M-bands (750 m spatial resolution). It is mounted on the NOAA-20 and NOAA-21 weather satellites of the joint Suomi-NPP mission.

The *Moderate Resolution Imaging Spectroradiometer* (MODIS) instrument was launched in 1999 with Terra and in 2002 with Aqua spacecrafts. It has a viewing swath width of 2,330 km and views the entire surface of the

Earth every one to two days. Its detectors measure 36 spectral bands and it acquires data at three spatial resolutions: 250 m, 500 m, and 1,000 m. The emissive bands at 1 km spatial resolution allow monitoring the land surface temperature and also the high temperatures associated with volcanic eruptions.

SkySat is a constellation of ~15 high-resolution Earth imaging satellites able to image revisit any location on Earth up to 10 times daily (a daily collection capacity of 4,000 km²/day.) *SkySat* images are approximately 50 centimeters per pixel resolution and are acquired in the RGB bands plus a near infrared channel. They are owned and operated by the commercial company Planet Labs¹.

PlanetScope is a constellation of approximately 130 satellites, able to image the entire land surface of the Earth every day (a daily collection capacity of 200 million km²/day) at approximately 3 meters per pixel resolution. Since the first launch in 2014, three *PlanetScope* instrument types have been released. The last one, the *SuperDove*, acquires data in the red, green, blue, near infrared, as well as a new red edge, green I, coastal blue, and yellow channels. *PlanetScope* is owned and operated by the commercial company Planet Labs¹.

The *Pléiades* satellite constellation (*Pléiades* 1 A and 1B, launched respectively in 2011 and 2012) delivers very high resolution optical imagery with a daily revisit capability for any point on Earth. The sensor acquires data in tri-stereo configuration allowing for the three-dimensional reconstruction of the Earth surface for a range of remote sensing applications in the fields of cartography, agriculture, forestry, hydrology and geological prospecting.

TROPospheric Monitoring Instrument (TROPOMI), which was launched in October 2017 aboard the ESA and Copernicus Sentinel-5 Precursor satellite, is specifically designed for atmospheric observations and provides a spatial resolution of 3.5 × 5.5 km² per pixel on the ground. The operational algorithm for SO₂ detection on *TROPOMI* uses the differential optical absorption spectroscopy (DOAS) method². The *TROPOMI* instrument captures data across four spectral regions (i.e., ultraviolet, visible, near-infrared, and short-wave infrared) enabling the detection of SO₂ along with several other gases (O₃, NO₂, CO, CH₄, and HCHO) and aerosols³.

Methods

Satellite data were promptly elaborated to monitor the eruptive activity of Etna during the 2025 February eruption. In particular, we processed SEVIRI, MODIS, VIIRS and FCI data using the CL-HOTSAT system to estimate the effusion rate and hence the lava flow volume, providing quantitative information about the different phases of the eruption. Moreover, we retrieved the thickness distribution of volcanic deposits by differencing successive digital topographies obtained by *Pléiades* imagery using the MicMac software. We also extracted the evolution of the lava flow field using *Skysat* and *PlanetScope* data, and estimated the sulfur dioxide total mass on Sentinel-5P *TROPOMI* imagery using a new clustering approach implemented in Google Earth Engine.

Details about the algorithms used, data processing and analyses, are reported below.

Estimation of the time-averaged discharge rate and erupted volumes. CL-HOTSAT is a satellite thermal monitoring system specifically designed for Mt Etna⁴, but successfully applied also to other volcanic areas^{5,6}. It ingests multi-spectral satellite data acquired by SEVIRI, MODIS and VIIRS sensors. CL-HOTSAT was recently specialized to process FCI data. For this sensor, we used the georeferencing datasets present in the NetCDF files, which provide the per row and per column satellite view angle in radians, limited to the specific chunk presented in the file. Once georeferencing information is converted into a form digestible by standard GIS software, and radiance information is obtained by calibrating the digital numbers, the script writes to disk a new raster, georeferenced through selected ground control points. This raster can be fed to the existing CL-HOTSAT system for further processing. The system uses the difference between the Middle Infrared and the Thermal Infrared bands and provides hotspot location in an automatic fashion. In the case of FCI we used the IR3.8 and the IR 10.5 bands. The hotspot algorithm is contextual and employs the well-known position of the volcanoes searching a threshold in a portion of the image considered “non volcanic”. For each hotspot pixel, the radiant heat flux is computed by following the MIR radiance approach⁷. This information can be converted into Time-Average Discharge Rate (TADR, i.e. the effusion rate averaged over given periods), from which the erupted volume is obtained by integration, in case of effusive eruptions.

Development of digital surface models and 3D mapping of volcanic deposits. To estimate the thickness distribution of the deposit and the lava volume emitted during the February 2025 Etna eruption, we used a topographic approach relying on the difference of *Pléiades*-derived DSMs. *Pléiades* images were processed using the free and open source MicMac photogrammetric library⁸. The pre-eruptive DSM was derived from a *Pléiades* triplet acquired on December 27th, 2023, while the post-eruptive DSM was obtained from the July 15th, 2024 and the March 6th, 2025, acquisitions. Since the March 6th acquisition was partially affected by cloud coverage, a fusion algorithm was applied to build a smoother homogeneous post-eruptive product⁹.

In order to reconstruct the 3D distribution of volcanic deposits minimizing the errors due to misalignments, we co-registered¹⁰ and then subtracted the two DSMs. The volume was calculated by integrating the thickness distribution over the area covered by the deposits. The uncertainty was quantified by multiplying the area and the standard deviation of terrain residuals outside of the deposits.

Tracking of lava flow emplacement. To ensure a detailed periodic mapping of lava flow advancement, we applied a semi-automatic thresholding and change detection algorithm, which has been successfully used during recent effusive eruptions at Etna^{11,12}. First, *Skysat* and *PlanetScope* images were imported into the GIS (specifically GRASS and QGIS) and RGB composites (using the RED, GREEN and near infrared spectral bands) were created to highlight the characteristics of the lava and detect changes associated with the emplacement of new lava among subsequent acquisitions. In this way, we obtained the boundaries of the lava flow on 9, 11, 12, 17, 21,

24, 26 February and 4 March (the final field). At first glance, we used Skysat images that have a higher resolution and therefore provide greater detail. In case of partial cloudiness or ash plume coverage, the Skysat images were complemented with Planetscope images acquired on the same day. GIS was also used to calculate the areas and perimeters.

Estimation of the sulfur dioxide total mass. Sulfur dioxide (SO₂) total mass was calculated on Sentinel-5P TROPOMI imagery, applying two different clustering techniques on the open-source Google Earth Engine computing platform. For the analysis, we exploited the offline Level 3 product at 15 km, applying a Simple Non-Iterative Clustering (SNIC) algorithm segmentation and then a *k*-means unsupervised classifier, followed by the application of a threshold for highlighting SO₂-contaminated pixels¹³. Specifically, we calculated the density for each of the classes derived from the previous clustering. The density is defined as the ratio between the total mass value of sulfur dioxide (obtained by summing the SO₂ values of each pixel within the class) and the relative number of pixels contained in that class. Ultimately, we selected only those classes with a density higher than 0.00028 mol/m², which provided an optimal value for the eruptive events occurring at Etna. This approach for object-based image analysis is designed to more accurately characterize sulfur dioxide, avoiding interference from noise that could derive, for example, from the sensor instrumentation onboard the satellites or from the lighting and atmospheric conditions¹⁴.

Data Records

The data record associated with this work consists of different products (Fig. 2), whose details about format and external links are provided below. All data are shared under the CC BY 4.0 use license and available at the link <https://doi.org/10.6084/m9.figshare.28759586>.

Time-averaged discharge rate and erupted volumes. We provide the TADR and volumes (with related uncertainties) extracted from SEVIRI, MODIS, FCI and VIIRS imagery from 8 February to 2 March 2025, which have been elaborated by the CL-HOTSAT system. We also provide location in terms of latitude and longitude for the hotspot pixels for each sensor. These estimates are supplied as an xlsx file with a tab for each sensor and one for the cumulative volumes (TADR&Volumes.xlsx). Date is expressed in Greenwich Mean Time (GMT) as dd/mm/yyyy hh.mm; minimum, mean and maximum value for TADR are expressed in m³/s; minimum, mean and maximum value for cumulative volumes in m³; latitude and longitude are expressed in decimal degrees.

Digital surface models and 3D mapping of volcanic deposits. We provide the most updated digital topography of Etna, obtained from Pléiades imagery acquired on March 6th, 2025 (Etna_DSM_25_5m.tif). The DSM covers an area of about 95 km², including the summit craters of Mt Etna and the Valle del Bove. It is provided as a Geotiff, with a spatial resolution of 5 meters. Coordinates refer to the WGS-84 ellipsoid; the map projection is UTM (Zone 33 North). The 3D mapping of volcanic deposits, which shows the thickness distribution of the final lava flow field, is provided as GEOTIFF (lava_flow_thickness.tif), with elevations in meters relative to the WGS-84 UTM 33 North Coordinate System (EPSG: 32633).

Lava flow emplacement. The actual fields of the lava flow emplaced during the February 2025 Etna eruption were retrieved for the following days: 9, 11, 12, 17, 21, 24, 26 February and 4 March. They are provided as vector shapefiles into the WGS84 UTM 33 North Coordinate System (EPSG: 32633), showing the areal development and the advancement of the lava front. The shapefile's attribute table includes fields called "area" and "perimeter" that contain the measurements of each map feature (lava_flow_fields.zip). The areas show an increment from 0.054 to ~0.94 km², which represents the total lava coverage.

Sulfur dioxide total mass. The retrieval of sulfur dioxide is provided in a csv file with the date expressed as dd/mm/yyyy and the minimum, mean and maximum total mass values in ktons (SO₂_Total Mass.csv). Some data are marked as "not available" ("N.A."), since they do not exceed the measurable detection limit by TROPOMI for SO₂, in terms of the lowest concentration that can be reliably detected (equal to 0.0124 ton).

Daily maps of pixels containing SO₂ are also provided in GeoTiff format (SO₂_maps.zip), which are georeferenced to the WGS-84 UTM 33 North coordinate system (EPSG: 32633).

Technical Validation

The height accuracy of the March 6th, 2025 DSM was estimated by using GPS Ground Control Points available on Etna volcano¹⁵. We performed a comparison between these points and the elevations of the DSM (excluding those falling inside volcanic deposits), finding residuals ranging between −3.6 and 2.8 m, with a mean value of 0.02 m and a standard deviation of 1.61 m, which represents the vertical accuracy of the DSM.

The DSM difference provides an estimation of the bulk lava volume, i.e. by including the vesicularity of the lava and possible cavities/tunnels, while CL-HOTSAT provides lava volume by integrating the TADR obtained from thermal multispectral images, i.e. based on the radiating portion of the volcanic deposits. By comparing the bulk volume from DSM difference ($5.05 \pm 0.9 \times 10^6$ m³) and the average volume from multispectral data ($5.37 \pm 0.8 \times 10^6$ m³), we found a difference of about 320,000 m³, which is well within the measurement errors⁷. The validation of the average volume from multispectral data was also done considering the estimates performed using drones, thermal cameras and field observations (INGV-OE, 2025a-d). These suggested an average volume of 0.25×10^6 m³ on February 11th, 2.1×10^6 m³ on February 16th, 3.5×10^6 m³ on February 24th and 4.5×10^6 m³ on March 2nd, with an uncertainty of 40%. Considering our satellite-derived estimates at 12:00 UTC of the same days, we obtain average volumes of 0.57×10^6 m³ on February 11th, 2.05×10^6 m³ on February 16th, 4.61×10^6 m³ on February 24th and 5.37×10^6 m³ on March 2nd.

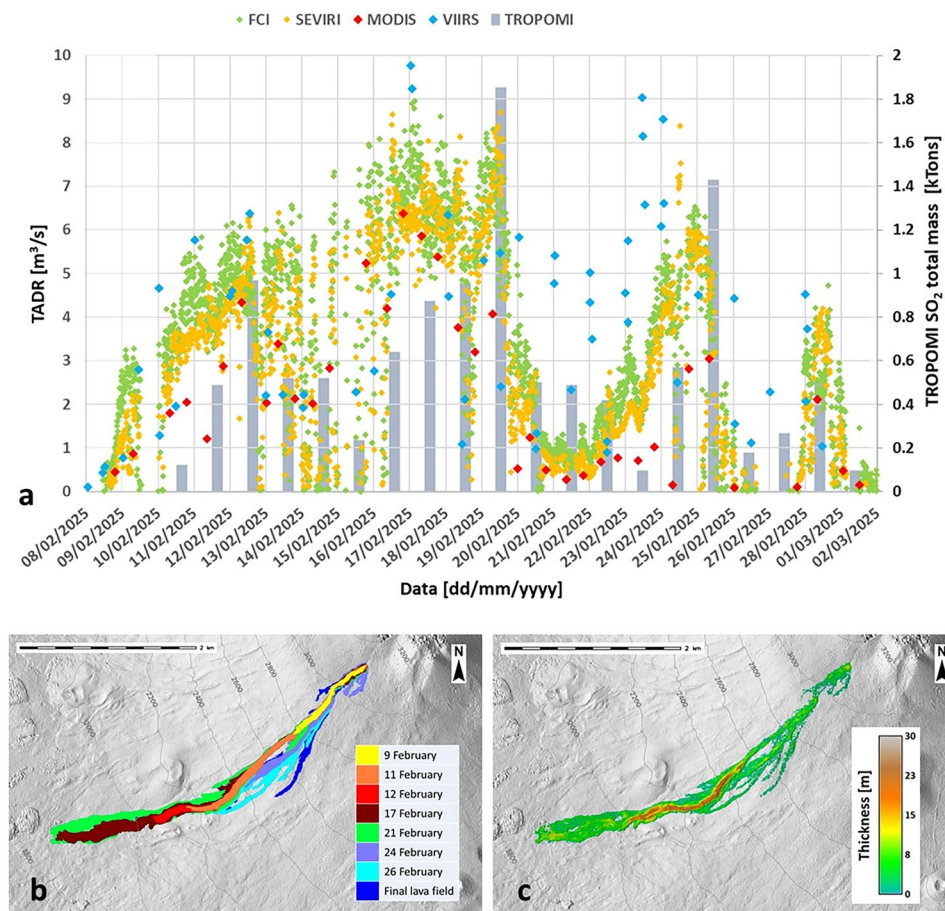


Fig. 2 The different satellite-derived products retrieved for the Etna 2025 eruption. (a) The TADR from SEVIRI (yellow diamonds), FCI (green diamonds), MODIS (red diamonds), and VIIRS (blue diamonds) data, together with the daily SO₂ total mass from TROPOMI (gray bars); (b) The evolution of the lava flow field from Skysat and PlanetScope imagery; (c) The volcanic deposits obtained by differencing Pléiades-derived DSMs.

The trend of the TADR estimated by CL-HOTSAT for the four different sensors is coherent and in line with the areal expansion of the lava flow field, with the maximum length reached just after the first TADR peak of February 16th. This was also confirmed by the field measurements made by the INGV personnel, who followed the advancement of the lava field and the minimum elevation reached^{16–18}.

The SO₂ mass flux is also in agreement with the TADR and in accordance with the SO₂ measured from the ground, which both present two peaks towards higher values on February 20th and 26th¹⁹.

The lava flow emplacement was validated using independent measurements obtained from drones, thermal cameras and field observations^{16–19}. In particular, we considered the maximum length of the main branch, the altitude reached by the front and the area covered by the lava. On February 11th, 16th, and 24th (extending to March 2nd), INGV-OE documented progressive lava flow lengths of 2.6 km, 4.2 km, and a peak of 4.3 km — reaching elevations of 2150, 1860, and 1840 m a.s.l. — with corresponding flow areas expanding from 1.3×10^5 m² to 3×10^5 m² and eventually stabilizing at 7×10^5 m². Our satellite-derived measurements indicate comparable or slightly greater flow lengths of 2.7 km (2160 m a.s.l.), 4.2 km (1860 m a.s.l.), and 4.3 km (1850 m a.s.l.), along with broader spatial extents of 1.6×10^5 m², 4.1×10^5 m², 7.6×10^5 m², and reaching 9.4×10^5 m² on March 2nd.

Code availability

No specific code was developed for this study.

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Author contributions

G.G.: Conceptualization. A.C., M.S., R.C., M.D., F.S., R.G., F.Z.: Visualization, Data Curation. G.G., A.C., G.B.: Investigation. G.B.: Formal analysis. G.G., G.B.: FCI, SEVIRI, MODIS data processing. G.G., F.S.: VIIRS data processing. M.D.: TROPOMI data processing. A.C.: Planet data processing. G.G.: Pléiades data processing. All authors contributed to writing and revising the manuscript, have approved the submission of the final version of the manuscript and take full responsibility for the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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